

Planting Material - Selecting Grape Varieties

Lesson Aim

Select and evaluate appropriate grape varieties for different situations.

Grape Types

Grapes are grown for a variety of crops, including table grapes, dried fruit, and to produce wine. Different varieties are suitable for different purposes. Grapes can generally be classified as true 'European varieties', 'American varieties' or 'French hybrids'.

European varieties are all derived from the species *Vitis vinifera*. These varieties are generally considered to be superior for winemaking. European varieties are also used in the production of table grapes, dried fruit and juice. European varieties are usually seeded grapes, although some seedless varieties do exist.

American varieties are cultivars of Native American species such as *Vitis labrusca* and *Vitis rotundifolia*. These species are native to the United States and have been used there since the eighteenth century. *Vitis labrusca* is the most commonly used American species in commercial viticulture. American varieties are generally considered to produce wine of inferior quality. They do not store well and are usually marketed locally. American varieties are, however, resistant to various pests and diseases, such as phylloxera. This resistance has led to their commercial usefulness.

French hybrids are the result of hybrids between European and American varieties. Opinions differ about the quality of French hybrids. Some authorities suggest that they produce inferior wines, others suggest that they can produce wine equal to those produced by European varieties of grapes. Proponents of French hybrids suggest that they can bring the best features of both the European and the American varieties together, providing vines that are resistant to significant pests or diseases, and produce high quality grapes.



Many vineyards, particularly in the northern hemisphere, utilise rootstocks to provide pest and disease resistance, as well as to modify the growth habits of vines. Rootstocks can provide growers with the capacity to manage pests such as phylloxera, whilst still producing specific varieties that may otherwise be unviable in a particular region. European varieties are commonly grafted onto resistant rootstocks originating from American varieties.

Other technologies that are utilised in varietal selection in viticulture include the certification of planting stock as 'free from known viruses', cloning of particular planting stocks, and genetic modification of planting stocks. Certification of plants as 'free from known viruses' involves subjecting plants to specific heat treatments to eradicate viruses, then producing planting stock from those 'clean' plants, for sale. Cloning involves producing genetically identical plants through vegetative propagation methods or tissue culture. Some grape varieties have been reproduced this way for centuries. Genetic modification involves altering the qualities of grapes (for example fruit size or disease resistance) by altering the vine's genetic material under laboratory conditions.

Selection Considerations

Various factors should be considered when selecting what grape varieties to use. Some particular factors include:

- Microclimate of the proposed vineyard. Which varieties will grow acceptably within this microclimate?
- Soil type of the proposed vineyard. Does the soil type lend itself to a particular variety of grape?
- Diseases present in the region. If there are particular diseases present in a region, then plants should be selected based on their disease resistance. Some regions may place restrictions on movement of planting stock due to disease problems (or freedom from disease problems).
- Rainfall of the proposed region. Are some varieties more or less suited to the rainfall volume and distribution of the region?
- Temperature. Are some varieties more or less suited to the temperature extremes, averages, and distribution in the region?
- Plant vigour. How vigorous do you require your vines to be? Different varieties and different combinations of rootstock and scion will perform differently in different regions.
- Yield. What yields can you expect from different varieties? Is yield quantity more likely to affect volume in some varieties?
- End use. Planting stocks should be selected based on the intended use of the grapes.

What To Plant

The varieties you choose to grow should be determined according to what you hope to use them for:

- Are they to be sold as fresh fruit (i.e. table grapes)?
- Are they to be dried (e.g. sultanas, raisins)? Drying generally requires a climate that has a low rainfall. (If rain gets on the skin of drying fruit it can crack and become diseased. If conditions are too humid during drying, fruit can become diseased.)
- Are they to be used for juice?
- Are they to be used for making wine or spirits and, if so, what type of product?

Table grapes need ample heat and a good water supply.

Wine grapes can be grown in a wider range of environments.

What Comes First: The Site or the Variety?

All too often, people first have the land and then look to choose a grape variety that suits their location. It is always better to start with the variety and then look for properties that have the conditions to grow that variety.

To ultimately achieve your goal of the best vineyard possible, it may in fact be a better proposition to sell a property which you already have, and buy a different property, in the same locality, but with a slightly different soil type or microclimate. Don't discount this option.

Matching the Variety with the Site

When choosing what variety to grow, you need to consider the following, and match the variety you choose with the conditions of the site:

- **Soil type.** Some grape varieties do well on a wide range of soils, while others need a particular type of soil to perform best.
- **Temperature and sunlight.** Temperature and sunlight play an important role in vine growth and development. Different varieties will vary in their temperature and sunlight requirements.
- **Rainfall.** Seasonal distribution of rainfall can improve or enhance different crop features. Analysing a region's rainfall can provide insight into factors such as irrigation requirements, disease risk, and varietal selection. Some varieties are more susceptible than others to wet weather at harvest. Some resist dry soil conditions a little better than others. Some need higher rainfall than others to perform well.
- **Pest and disease susceptibility.** Some varieties are far more likely to suffer from pests or diseases. Pest or disease-resistant varieties may suit one climate but not another. Some disease-prone varieties may yield desirable grapes, and techniques such as grafting can facilitate growing these varieties in disease-prone areas. Some geographical locations place restrictions on movement of vine stock, in order to prevent pest and disease infection.
- **Vigour.** Vigour can vary also. Some need to be treated a particular way to maintain reasonable vigour; while others are relatively vigorous irrespective of how they are treated. In some conditions, having vines of lower vigour may be desirable.
- **Yield.** Like everything else this can vary from low to high, depending on variety. Some wine experts believe that higher yields tend to reduce grape quality. This may be truer of some varieties than others. Different kinds of yields are possible in different climates because climate has an impact on yield and on the kinds of grape production possible.
- **Rootstocks.** Rootstocks are widely used to improve disease and pest resistance and enhance growth and cropping characteristics. However, rootstocks must be selected carefully to ensure satisfactory growth and fruit ripening. This requires matching the rootstock not only with the grafting material but also with the climate and soil type.

Varietal Characteristics

Appropriate varietal selection is paramount for successful yields and the production of high quality fruit. Time and effort invested in site selection and preparation and vine management will not be rewarded if the correct grape varieties to suit the growing conditions are not made. This is particularly true in marginal growing regions such as many parts of the UK. There are numerous cool climate varieties with a range of growth characteristics.

Major characteristics of grape vines include vigour, rootstock types, best pruning methods, and nature of shoots. Cabernet Sauvignon is a vigorous variety with low yields. Merlot grows less vigorously, yields higher, but tends to have a late bud burst so can be more frost hardy. Pinot Noir grapes produce excellent, flavourful wines which can have a very full-bodied, complex nature; however, this variety grows best in deep, well-drained calcareous soils. Riesling is another variety that can be productive in cold climates but can be compromised by fertile, loamy soils as are often found in the UK.

If the home grower has no ambition making wine from their grapes, then other varieties would be better chosen. For example, again using the example of the UK grower, white grape varieties could yield better than red or black varieties which might not fully develop and or might only produce small berries. Varieties such as Bacchus and Orion mature in mid-autumn and have good disease resistance.

Selecting Wine Grapes

When selecting what to grow, the prime concern should be the varieties' suitability to your locality. Large vineyards can experiment, but small operations must consider their selection very carefully.

Many wineries today do not grow all of their own grapes. They often buy grapes from growers in their district or surrounding districts. If you locate a new vineyard in a region that already has large well-established wineries doing this, you are enhancing the prospect for sales. Even if you do have a market for your produce, there may be occasional boom years, or years when market conditions change, and to have the possibility of selling to other wineries, not too far away, can be a big advantage. Similarly, if you cannot produce sufficient grapes for your own needs one year, you may still have a possibility of buying grapes from other vineyards in your locality.

Consider:

Chardonnay: ripens early, and is not normally damaged by rain.

Chenin Blanc: ripens early to mid-season; is susceptible to splitting by rain and to bunch rot, and should be grown only in warm areas.

You should make comparisons like this between all the varieties you are considering.

Yield

The yield per acre should be considered in choosing what varieties to plant. This will affect the economic viability of your operation. The following varieties will likely perform as outlined below in an area with low to medium rainfall and good soil.

Variety	Tonnes Per Acre in 475 mm rainfall	Tonnes Per Acre in 575 mm rainfall
Cabernet Sauvignon	2 - 5	6 - 11
Shiraz	3 - 6	7 - 14
Grenache	4 - 10	8 - 16
Rhine Riesling	2 - 5	6 - 11

2 tonnes per acre would be considered low yield.

10 tonnes per acre would be considered high yield.

Low yield varieties include Cabernet Sauvignon, Gewürztraminer and Riesling.

Low to medium yield varieties include Chardonnay, Pinot Noir, Rieslaner and Sauvignon Blanc.

Medium yield varieties include Ehrenfeleser, Malbec, Merlot, Meunier, Muscat, Perle, Pinot-gris, Reichensteiner and Semillon.

Medium to high yield varieties include Chasselas, Chenin Blanc, Faber, Kerner, Malbec, Pinot Blanc, some new Riesling cultivars, Siegerebe and Sylvaner.

High yield varieties include Bacchus, Kerner, Muller-Thurgau, Mario Muscat and Zweigeltrebe.

Note the indication of yields above is a generalisation. The locality they are grown in can also strongly affect the yield. Mueller-Thurgau, for example, is a low yield variety in Britain, but high yield in some other localities.

Wine Grape Varieties**Chenin Blanc**

A premium white wine in France. A quality wine in the dry, sweet or sparkling style. Late harvest grapes.

Characteristics: Produces a light wine that retains a pronounced varietal character with medium acidity. Late harvest grapes give a distinctive wine with complex character.

Culture

- **Soils** - Calcareous soils are said to improve quality considerably in France, although this has not been confirmed in South Africa or California where it appears less demanding in relation to the soil.
- **Vigour** - Vigorous.
- **Pruning** - Medium to long canes with a high proportion of old wood to new canes (cordons).

- **Disease susceptibility** - Downy mildew - medium to high, Powdery mildew - moderate to high, Bunch rot and *Botrytis* - moderate to high.
- **Wet weather** – Susceptible.
- **Rootstocks** - Use those with moderate vigour.
- **Yields** - Moderate to high.

Chardonnay (Pinot Chardonnay)

A premium quality white wine variety of central Europe. Has a distinctive wide open, basal sinus and yellow-green grapes. In soils lacking calcium the wine is less aromatic.

Characteristics: The wine has a fine and distinctive varietal aroma and flavour.

Culture

- **Soils** - Likes well drained, fertile but drier calcareous soils. Not suited to heavy clay loams. Heavy nitrogen fertilisers not recommended since set becomes poor and grapes are more susceptible to rots.
- **Vigour** – Moderate.
- **Pruning** - Medium to long cane pruning.
- **Disease susceptibility** - Downy mildew - low, Powdery mildew - fair, Bunch rot and *Botrytis* - moderate to high.
- **Wet weather** - Susceptible to rain at harvest.
- **Rootstocks** - Vigorous ones preferred.
- **Yields** - Low to moderate - depending on clone and climate.

Pinot Blanc

A premium white French variety. Early ripening grapes.

Characteristics: The wine retains fair acidity with fine varietal flavour and aroma.

Culture

- **Soils** - Likes well drained stony or calcareous soils. It has resistance to drought.
- **Vigour** – Moderate.
- **Pruning** - Use long canes.
- **Disease susceptibility** - Downy mildew - low, Powdery mildew - fair to high, Bunch rot and *Botrytis* - fair to high.
- **Wet weather** - Fair resistance.
- **Root stocks** - Does well on vigorous root stocks.
- **Yields** - Moderate to high.

Semillon

A white grape grown throughout France for its fine and dry or late harvested sweet wines of considerable finesse and aroma. In Australia it is known as Hunter Riesling and White Madeira; in South Africa it is known as Green Grape.

Characteristics: Soft, fair acidity in cool climates, and a distinctive herbaceous aroma and flavour.

Culture

- **Soils** - Prefers lighter soils.
- **Vigour** - Moderate to vigorous.
- **Pruning** - Use long canes and summer prune in more humid areas.
- **Disease susceptibility** - Downy mildew - fair, Powdery mildew - fair to high, Bunch rot and *Botrytis* - high, but see comments under Sauvignon Blanc.
- **Wet weather** - Poor resistance, also susceptible to sunburn.
- **Yields** - Moderate.

Muscat Ottonel

A quality white grape grown throughout Europe. Early ripening. Short internodes are typical.

Characteristics: Low acidity, fine aroma, pronounced varietal flavour.

Culture

- **Soils** - Deep fertile soils are best; calcareous soils induce chlorosis. Drought conditions produce insufficient growth.
- **Vigour** - Low to moderate with a tendency to spread rather than grow upright, therefore a smaller system such as head cane are recommended.
- **Pruning** - Use short canes.
- **Disease susceptibility** - All diseases - fair.
- **Wet weather** - Moderately susceptible.
- **Rootstocks** - Moderate to vigorous. Choose those like 26-G or SO-4 which improve set: an inherent problem with this variety.
- **Yields** - Moderate, but consistent in cropping.

Muscadelle

A widely-planted white grape variety throughout the world. Young spring growth is reddish. Fruit ripens early.

Characteristic: Marked bouquet somewhat reminiscent of Muscat.

Culture

- **Soils** - Well drained, deep profiled soil.
- **Vigour** - Fairly vigorous.
- **Pruning** - Use medium canes.
- **Disease susceptibility** - All diseases - moderate.
- **Wet weather** - Does not appreciate hot humid weather.
- **Yields** - Moderate.

Gewurztraminer (Savagnin Rosé)

Produce both white and pink berries in the one bunch. Matures early. Leaf is distinctive, being small, round, dark green and rough with shallow lobes. This is a clonal selection of the Traminer variety.

Characteristics: Distinctive spicy wine, pronounced varietal aroma and flavour, and low acidity.

Culture

- **Soils** - Likes deep fertile soils, use nitrogen with caution since excess causes poor set. Has a tendency to exhibit chlorosis in calcareous soils.
- **Vigour** - Moderate, has short internodes, lacks upright growth. There is a tendency for multiple shoots to be produced from the buds.
- **Pruning** - Use long canes with a higher proportion of older wood in a cordon system. Summer thinning of the shoots is advisable.
- **Disease susceptibility** - Downy mildew - fair, Powdery mildew - high, Bunch rot and *Botrytis* - high.
- **Wet weather** - Susceptible to wet weather at flowering and ripening, causing poor set and splitting of berries.
- **Rootstocks** - Low to moderate vigour preferred.
- **Yields** - Low to moderate.

Cabernet Sauvignon

A premium black grape. In cool climates it ripens late in the season, giving small to medium yields of high quality fruit. This grape will not ripen fruit in very cool districts. The quality of the wine compensates for the generally low yields. Leaf is distinctively dark green and rough on the surface with deep sinuses; reddish shoot growth in spring is typical.

Characteristics: Fullness of varietal, depth in bouquet and flavour, coupled with high acidity and tannin content.

Culture

- **Soils** - Suits most soils.
- **Vigour** – Vigorous.
- **Pruning** - Medium to long cane pruning preferred with a high proportion of old wood to young canes (use cordon).
- **Disease susceptibility** - Downy mildew - fair, Powdery mildew - high, Bunch rot and *Botrytis* - low.
- **Wet weather** - Good resistance to wet weather at harvest.
- **Rootstocks** - Low to moderate vigour are best to control vigour and improve the set.
- **Yield** - Low, but can be increased with good pruning.

Carignan

One of the most widely grown red wine grapes in the world. Has large lobed leaves with hairs on the underside. Fruit matures late.

Characteristics: Moderate colour, good tannin and no pronounced varietal character.

Culture

- **Soils** - Sandy soils to most soil types.
- **Vigour** - Vigorous and upright.
- **Pruning** - May need summer pruning.
- **Disease susceptibility** - All diseases - low.
- **Wet weather** - Fruit not readily damaged by rain during ripening.
- **Yields** - Up to 20 to 30 tonnes per hectare is possible.

Vitis rotundifolia

This is an American species, commonly known as Muscadines, Scuppernongs or “Scupps”. Scupps differ to *Vitis vinifera* in a number of ways:

- Highly resistant to disease
- Larger, looser bunches
- Berries can be larger (to 50mm diameter)
- Berries in a bunch ripen progressively (not all at once)
- Berry colour is much more variable than *V. vinifera*
- Berry skin is loose
- Taste is different to *V. vinifera* (some people say a little like pineapple flavour)
- Plants are either male or female

Cultivated named varieties of Scupps are available in nurseries throughout North America, but not everywhere else in the world. In many countries (e.g. Australia) it is difficult to find cultivars of this species, and even importing them can be a problem due to quarantine regulations.

Importing seed may be easier than importing plants, but the reliability of seedlings to provide desirable fruit is uncertain. You may need to grow and select the best plants, if attempting to establish *V. rotundifolia* plantings this way.

Wine Grapes

For dry or table wines, high acidity and moderate sugar content are desired. Grapes with high sugar content and moderately low acid are required sweet or desert wines. Grapes such as Cabernet Sauvignon, Riesling and Pinot Noir all have outstanding bouquet and flavour essential for production of the highest quality premium wines. Thin-skinned varieties will require more care during harvest and post-harvest techniques than those with thick skin and a firmer pulp.

Raisin Grapes

Dried raisins must be soft in texture, should not stick together when stored, and have good flavour. Early ripening is desirable so that the grapes can be dried rapidly during favourable weather. Seedlessness is preferred. Large berries are preferred for eating; small berries are preferred for the bakery trade.

Some of the best varieties for raisin production are Thompson Seedless, Black Corinth, and Muscat of Alexandria (which has seeds that can be removed by machine).

Juice Grapes

The natural flavour of the grapes should be destroyed by the clarifying and preserving procedures. In the United States juice is usually pasteurised after it is clarified thereby destroying the natural flavour. In the US juice is usually produced from Concord grapes or blends of Concord with other grapes.

The Importance of Rootstocks

Grape vines are commonly (but not always) grown as budded or grafted varieties on a rootstock (i.e. the root system is one variety of plant and the top (scion) a different variety). In some respects the choice of an appropriate rootstock can be just as important as the choice of the variety on top which produces the fruit.

Rootstocks may contribute to the top in a variety of ways, including:

- Providing resistance from phylloxera;
- Making fruit mature earlier;
- Providing overall increased vigour of growth;
- Improving tolerance to saline soil;
- Resisting nematode damage to roots.

Purchasing Plants

It is wise to buy established plants rather than trying to propagate your own. The productivity of your vineyard will depend – perhaps more than anything else – on the quality of the plants. Aside from this, it will generally take you more than a year to propagate plants that could otherwise have been bought at a reasonable price.

You must decide whether to buy rootlings – cuttings, or grafted plants. Obviously grafted plants are better, but in a disease-free area they may not be necessary. The supplier should also be able to offer you virus-free stock. This may be worth considering.

Check prices and quality with several different suppliers. Don't buy from the first nursery you visit. Generally, there are vine propagation nurseries to be found in vineyard areas. Check in the telephone directories for those areas.

Some varieties may be more difficult to obtain than others.

Phylloxera

Grape Phylloxera is a small aphid which lives on the roots (and occasionally the leaves) of grapevines. It is native to the eastern parts of North America, but during the 1800's was spread to grape growing areas throughout the world.

It has been a significant problem not only in North America, but also Australia and other parts of the world.

Phylloxera aphids cause galls or swellings on the roots of vines which eventually decay. This process gradually reduces the vigour of the vine, and over three to ten years a healthy productive vine can turn into an unhealthy, unproductive plant.



Phylloxera can be devastating to a vineyard, and once infested it is difficult to bring under total control. The best method of control is to graft plants onto phylloxera-resistant root stocks. Such grafted vines produce consistent and reliable crops, even when phylloxera is present.

Rootstock varieties which have a high resistance to phylloxera include:

Ramsey, Schwarzmann, 5A Teleki, 99 Richter, 100 Richter, 5BB Kober, Freedom, 140 Ruggeri and 34 EM.

Ramsey is best suited to hot to warm areas and sandy soils.

Schwarzmann grows in any type of soil and most climates.

5A Teleki and 110 Richter grow on most soil types but only in warmer climates.

5BB Kober grows on most soils but prefers milder climates, certainly not hot areas.

Quarantine is used in some places to prevent phylloxera spreading into unaffected vineyard areas. Because the aphids only occur on grapevines, they are relatively easy to control by quarantine if policed properly amongst new plantings. It is most important that you always use phylloxera-free and phylloxera-resistant plants when planting vines.

- Never take vines or any part of a vine from a phylloxera-affected area.
- Never take soil from vineyards in a phylloxera-affected area.
- Do not grow cuttings or grafts taken from a phylloxera-affected area.
- Do not enter a vineyard without the owner's knowledge or permission.

Areas in Victoria, Australia, where phylloxera still exists, are called Vine Disease Districts. It is illegal to remove grapevines, or any part of the grapevine, from a Vine Disease District.

CHARACTERISTICS OF COMMONLY USED ROOTSTOCKS									
	Phylloxera resistance		Nematode resistance			Scion vigour	Lime tolerance	Drought tolerance	Comments
	Root	Leaf	Rootknot	Dagger	Root lesion				
Ramsey	3	1	3	1	3*	3	2	3	Hot to warm areas, sandy soils
Schwarzmann	3	1	3	3	-	2	2	1	Hot to cool areas, range soils
5A Teleki	3	1	2	-	-	2	3	2	Hot to warm areas, range soils
SO4	2	1	2	2	-	2	2	2	Warm to cool areas, range soils
99 Richter	3	1	2*	2*	2	2	2	3	Warm to cool areas, range soils*
K51-40	2*	2*	3	-	-	3	1*	1	Hot to warm areas, sandy soils
K51-32	2	1	3	2	-	3	1	1	Hot to warm areas, sandy soils
5BB Kober	3	1	2	2*	-	2	2	2	Warm to cool areas, range soils
Dog Ridge	2*	2	3	3	3*	4	2	2	Hot areas, sandy soils
Freedom	3	-	3	1	-	3	-	-	Hot to cool areas, range soils*
Harmony	1	1	2	3	2	3	2	1	Hot to warm, sand to loam soils
140 Ruggeri	3	-	2	1*	-	2	3	4	Hot to cool areas, range soils
110 Richter	3	3	2	1	1	1	2	4	Hot to warm areas, range soils
34EM	3	-	2	2	-	2	2	1	Hot to warm areas, range soils
Phylloxera resistance of other rootstocks: High – Rupestris du Lot, Riparia gloire, 101 – 14, 3306, 3309, 420A, 1103P, 106.8. Moderate – 1613, 1202. Low – ARG No. 1 None – V. vinifera									
Resistance rating:	1 = low	2 = moderate	3 = high	4 = very high	- = unknown				
Further information required in Australia: *									

